Training Day 2

1. Topics Covered and Brief Explanation

1. Introduction to Machine Learning

Machine Learning is a subset of Artificial Intelligence that enables computers to learn from data and improve their performance without being explicitly programmed for every individual task.

2. Working Principles of Machine Learning

Studied how machine learning systems use data to identify patterns, learn relationships and make predictions or decisions. Understood the basic workflow of collecting data, training a model and generating predictions.

3. Relationship Between AI and ML

Learned that Artificial Intelligence is the broader field of creating intelligent systems, whereas Machine Learning is one of the approaches used to achieve AI.

4. Importance of Machine Learning

Studied how ML helps in:

- Automating repetitive tasks.
- Predicting future outcomes.
- Identifying patterns in large datasets.
- Improving decision-making.
- Providing personalized recommendations.

5. Real-World Applications of Machine Learning

Explored applications such as:

- Email spam detection.
- Movie and product recommendations.
- Fraud detection in banking.
- Weather forecasting.
- Speech recognition.
- Customer behaviour prediction.

2. Tools and Technologies Used

Tool	Purpose
Google Chrome	Researching machine learning concepts.

Google Search	Exploring ML applications and examples.
YouTube	Learning the basic working principles of ML through tutorials.
Google Colab	Exploring an online environment for Python-based ML experiments.

3. Practical Learning / Activities

- Explored examples of machine learning in everyday applications.
- Studied how email spam filters identify unwanted messages.
- Understood the basic process of training a machine learning model using sample data.
- Compared traditional programming with machine learning.

4. Learning Outcomes

- Understood the fundamental concepts of Machine Learning.
- Learned how machines identify patterns from data.
- Understood the relationship between Artificial Intelligence and Machine Learning.
- Identified different applications and benefits of ML in real-world scenarios.